AI-Assisted Boolean Expression Simplifier

Digital Electronics Lab (ESC392) — Part B, Assignment 1

1. GenAI Model Name

ChatGPT (OpenAI)

2. Prompt

“Simplify the four-variable Boolean expression F(A,B,C,D) = Σm(0,1,2,3,8,9,10,11) using a Karnaugh map. Show the simplification steps, provide the original and simplified Boolean expressions, and explain how to implement and verify both circuits in Logisim Evolution.”

3. Response

The given Boolean expression is:

F(A,B,C,D) = Σm(0,1,2,3,8,9,10,11)

The minterms are grouped in a four-variable Karnaugh map. The eight 1s form a group of eight using the wraparound property. In this group, A, C, and D change, while B remains 0.

Therefore, the simplified Boolean expression is:

$$F = \overline{B}$$

The original circuit can be implemented using NOT gates, eight 4-input AND gates, and an OR gate. The simplified circuit requires only one NOT gate connected to input B.

4. Verification by Logisim

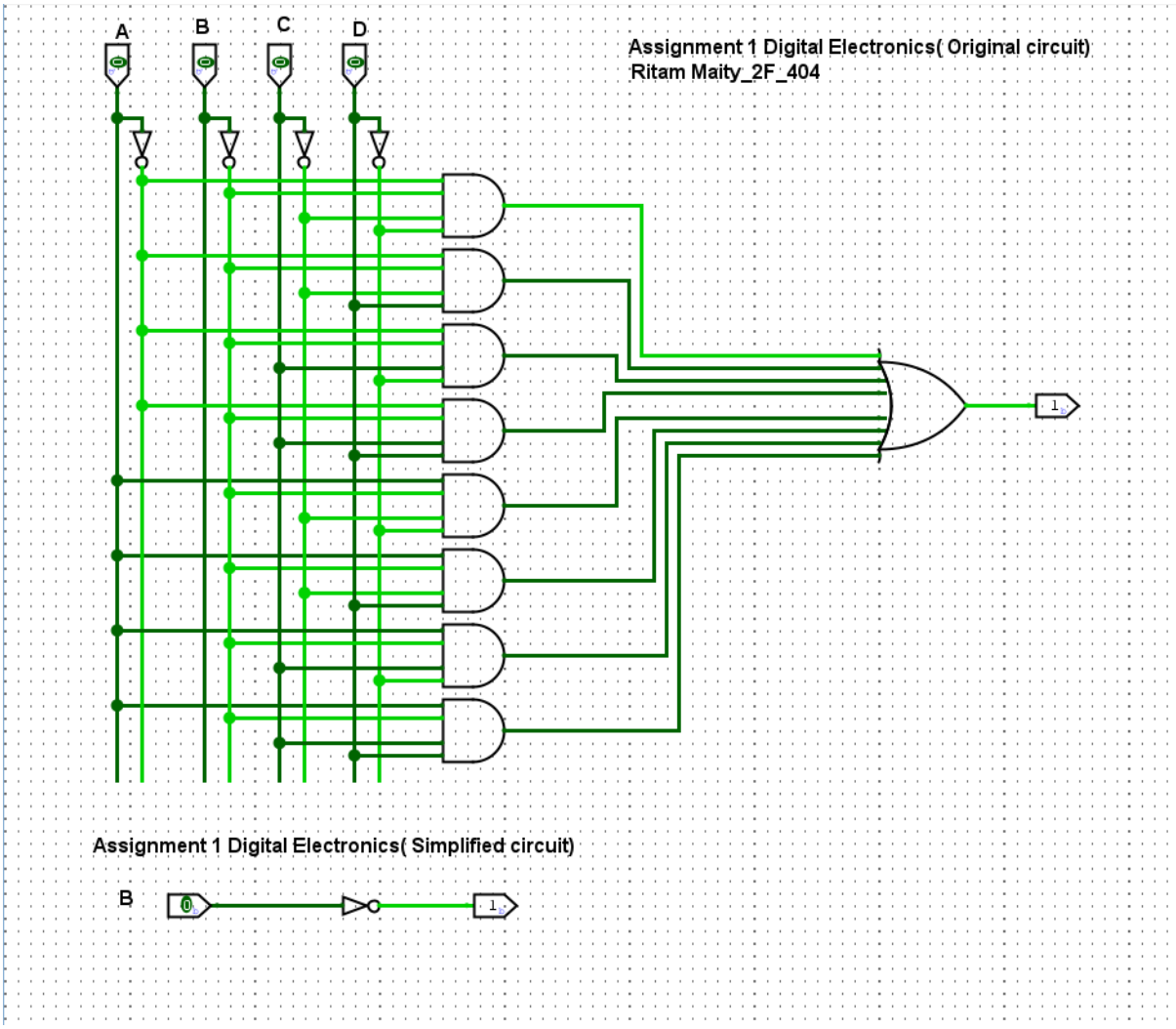
Both the original and simplified circuits are implemented in Logisim Evolution. All 16 possible input combinations of A, B, C, and D are tested. The outputs of both circuits are compared for every combination.

Expected result: The original and simplified circuits produce identical outputs for all 16 input combinations.

5. Circuit (Ckt)

Original circuit: Implement the canonical SOP expression using NOT, AND, and OR gates.

Simplified circuit: Connect input B to a NOT gate and connect its output to F.



6. Output

The original and simplified circuit outputs match for all 16 input combinations. The simplified function is $F = \overline{B}$.

7. Discussion

The four-variable Boolean expression was simplified using a Karnaugh map. The simplified expression requires only one NOT gate, reducing the circuit complexity. The truth table confirms that the original and simplified expressions are logically equivalent. Logisim simulation screenshots will provide evidence of the practical verification.

8. GitHub Link

GitHub Repository URL: https://github.com/Ritam-Maity/Digital-Electronics-Lab

The repository contains the Logisim circuit files and simulation screenshots for this assignment.